ABHIJIT SHINDE

RF & Microwave Engineer — PhD Candidate, Electrical & Computer Engineering

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RF and microwave engineer and ECE PhD candidate with hands-on experience across the full RF development cycle — antenna design, electromagnetic simulation, fabrication, and VNA-based bench validation. Skilled in Ansys HFSS, CST Studio Suite, Keysight ADS, and XFdtd. Seeking a Summer 2027 internship in RF/microwave engineering, antenna design, or wireless communications. Eligible for CPT work authorization (F-1).

EDUCATION

Ph.D., Electrical & Computer Engineering, North Carolina A&T State University | Aug 2023 – Dec 2027 (expected) | GPA 3.68/4.0

M.E., Electronics & Telecommunication Engineering, Savitribai Phule Pune University | 2016

B.E., Electronics & Telecommunication Engineering, Shivaji University | 2007

Graduate coursework: Microwave Circuits · Semiconductor Devices · Estimation Theory · Linear Control Systems · Network Optimization

TECHNICAL SKILLS

EM / RF simulation: Ansys HFSS (Electronics Desktop), CST Studio Suite, Keysight ADS, XFdtd

RF design: S-parameters, impedance matching, Smith chart, transmission-line & microstrip modeling; filter design (YIG / tunable / microstrip), couplers, power dividers, matching networks, beamforming

Antennas: Vivaldi (UWB), patch, slot, planar arrays; mutual-coupling analysis; SAR / near-field

Measurement & prototyping: VNA calibration (SOLT / TRL), return loss / VSWR / gain, anechoic-chamber pattern measurement, PCB prototyping & RF assembly

Programming: MATLAB, Python (RF data post-processing, test automation)

RESEARCH EXPERIENCE

Ph.D. Researcher — RF, Microwave & Antenna Engineering North Carolina A&T State University | Aug 2023 – Present

- Designed, fabricated, and bench-tested UWB Vivaldi antenna arrays in Ansys HFSS and XFdtd for wideband (1–7 GHz) microwave imaging; achieved |S11| < −10 dB and ~4 dBi realized gain, with reduced mutual coupling through array and feed optimization.
- Developed a planar YIG-based magnetically tunable microwave bandpass filter for frequency-agile RF front ends; presented at IEEE AP-S/URSI 2026.
- Performed S-parameter analysis (S11/S21), impedance matching, and SAR / near-field evaluation; correlated measured vs. simulated results to account for fabrication tolerances.
- Built an automated microwave scanning platform (LiteVNA 64, Arduino, MATLAB) for S-parameter data acquisition.
- Currently working on direct antenna modulation (BPSK and QPSK).
- Developed CNN-based signal-processing pipelines classifying S-parameter data with >95% accuracy for intracranial hemorrhage detection.

Graduate coursework projects North Carolina A&T State University

- Multi-stage broadband impedance-matching network design for microwave circuits; microstrip low-pass filter design with simulation-based validation.
- Dynamic spectrum management and adaptive channel selection for cognitive radio networks.

TEACHING & INDUSTRY EXPERIENCE

Assistant Professor, Electrical Engineering — Sinhgad Institute of Technology (Jul 2022 - May 2023); Dr. D. Y. Patil Institute of Technology (Aug 2017 - Jul 2022). Taught electromagnetics, DSP, image processing, and AI; supervised undergraduate antenna and signal-processing projects.

Automation Engineer — KalpTech Solutions (Jan 2014 - Jan 2016); Rapid Pack Engineering (Feb 2008 - Oct 2009). Built ML-based machine-vision inspection systems; programmed industrial control systems.

PUBLICATIONS & RECOGNITION

- 5 peer-reviewed publications; 10 citations. Presented “Design of a Planar YIG-Tunable Microwave Bandpass Filter” at IEEE AP-S/URSI 2026.
- IEEE Antennas & Propagation Society — Graduate Student Member.
- NASA MUREP reviewer · Best Research Paper Award, ICIC conference · Class of 1969 Endowed Scholarship, NC A&T (2026–27).